

The Exponential Function

Complex Analysis

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The Existence Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness Theorem

The exp Function

The Trigonometric Functions

The number π

The Hyperbolic Functions

Euler's Formula

The Polar Representation of $\mathbb{C}$

De Moivre's Formula

The n th Roots of Unity

Theorem

There exists a function $E : \mathbb{R} \rightarrow \mathbb{R}$ such that:

- i) $E'(x) = E(x)$ for all $x \in \mathbb{R}$.
- ii) $E(0) = 1$.

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Corollary

The function E has a derivative of every order and $E^{(n)}(x) = E(x)$ for all $n \in \mathbb{N}, x \in \mathbb{R}$.

Definition

If a function f has an n th derivative at a point $x_0 \in \mathbb{R}$, it is not difficult to construct an n th degree polynomial P_n such that $P_n(x_0) = f(x_0)$ and $P_n^{(k)}(x_0) = f^{(k)}(x_0)$ for $k = 1, 2, \dots, n$. In fact, the polynomial

$$P_n(x) := f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

has the property that it and its derivatives up to order n agree with the function f and its derivatives up to order n at the specified point x_0 . This polynomial P_n is called the **n th Taylor polynomial for f at x_0** . It is natural to expect this polynomial to provide a reasonable approximation to f for points near x_0 , but to gauge the quality of the approximation, it is necessary to have information concerning the remainder $R_n := f - P_n$. The following fundamental result provides such information.

Theorem (Taylor's Theorem)

Let $n \in \mathbb{N}$, let $I := [a, b]$, and let $f : I \rightarrow \mathbb{R}$ be such that f and its derivatives $f', f'' \dots, f^{(n)}$ are continuous on I and that $f^{(n+1)}$ exists on (a, b) . If $x_0 \in I$, then for any $x \in I$ there exists a point c between x and x_0 such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}.$$

The formula

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

is referred to as the **Lagrange form** (or the **derivative form**) of the remainder.

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Theorem

The function $E : \mathbb{R} \rightarrow \mathbb{R}$ that satisfies i) and ii) of the previous theorem is unique.

Theorem

The exponential function satisfies the following properties:

- iii) $E(x) \neq 0$ for all $x \in \mathbb{R}$;
- iv) $E(x + y) = E(x)E(y)$ for all $x, y \in \mathbb{R}$;
- v) $E(rx) = (E(x))^r$ for all $x \in \mathbb{R}, r \in \mathbb{Q}$.

Definition

The unique function $E : \mathbb{R} \rightarrow \mathbb{R}$, such that $E'(x) = E(x)$ for all $x \in \mathbb{R}$ and $E(0) = 1$, is called the **exponential function**. The number $e := E(1)$ is called **Euler's number**. We will frequently write

$$\exp x := E(x) \quad \text{or} \quad e^x := E(x) \quad \text{for} \quad x \in \mathbb{R}.$$

Definition

The function defined by the series

$$E(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

*converges uniformly on $\mathbb{C}$, absolutely for all $z \in \mathbb{C}$ and is called the **complex exponential function**[†]. We will frequently write*

$$\exp z := E(z) \quad \text{or} \quad e^z := E(z) \quad \text{for} \quad z \in \mathbb{C}.$$

[†]Since we will be working mostly on the complex realm, we may refer to the complex exponential function simply as the exponential function.

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Theorem

The complex exponential function is unique for each $z \in \mathbb{C}$ and satisfies the extension of properties i), ii), iii), and iv).

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Example

Show that the modulus of e^z is $e^{\Re(z)}$. That is, the modulus of the exponential function is only determined by the real part of the argument.

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Theorem

There exist functions $C : \mathbb{R} \rightarrow \mathbb{R}$ and $S : \mathbb{R} \rightarrow \mathbb{R}$ such that

- i) $C''(x) = -C(x)$ and $S''(x) = -S(x)$ for all $x \in \mathbb{R}$.
- ii) $C(0) = 1$, $C'(0) = 0$, and $S(0) = 0$, $S'(0) = 1$.

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Corollary

If C, S are the functions from the previous theorem, then

iii) $C'(x) = -S(x)$ and $S'(x) = C(x)$ for $x \in \mathbb{R}$.

Moreover, these functions have derivatives of all orders.

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Corollary

*The functions C and S satisfy the **Pythagorean Identity**:*

iv) $(C(x))^2 + (S(x))^2 = 1$ for $x \in \mathbb{R}$.

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Theorem

The functions C and S satisfying properties i) and ii) are unique.

Definition

The unique functions $C : \mathbb{R} \rightarrow \mathbb{R}$ and $S : \mathbb{R} \rightarrow \mathbb{R}$ such that $C''(x) = -C(x)$ and $S''(x) = -S(x)$ for all $x \in \mathbb{R}$ and $C(0) = 1, C'(0) = 1$, and $S(0) = 0, S'(0) = 1$, are called the **cosine function** and the **sine function**, respectively. We ordinarily write

$$\cos x := C(x) \quad \text{and} \quad \sin x := S(x) \quad \text{for} \quad x \in \mathbb{R}.$$

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Lemma

There exists a root ϖ of the cosine function in the interval $(\sqrt{2}, \sqrt{3})$. Moreover $C(x) > 0$ for $x \in [0, \varpi)$. The number 2ϖ is the smallest positive root of S .

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Definition

Let $\pi := 2\varpi$ denote the smallest positive root of S . We will often refer to π as the **number pi**

Definition

The unique functions $c : \mathbb{R} \rightarrow \mathbb{R}$ and $s : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

- i) $c''(x) = c(x)$ and $s''(x) = -s(x)$ for all $x \in \mathbb{R}$;
- ii) $c(0) = 1, c'(0) = 0$ and $s(0) = 0, s'(0) = 1$;

*are called the **hyperbolic cosine** and **hyperbolic sine functions**, respectively.*

The hyperbolic cosine and hyperbolic sine functions are commonly denoted by $\cosh$ and $\sinh$, respectively.

Definition

We define the extensions of the sine, hyperbolic sine, cosine and hyperbolic cosine functions by

$$\sin z = \frac{1}{2i} (e^{iz} - e^{-iz}),$$

$$\cos z = \frac{1}{2} (e^{iz} + e^{-iz}),$$

$$\sinh z = \frac{e^z - e^{-z}}{2},$$

$$\cosh z = \frac{e^z + e^{-z}}{2}.$$

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Exercise

Find a power series representation for $\sin z$ and $\cos z$.

Proposition

The functions $\sin : \mathbb{C} \rightarrow \mathbb{C}$ and $\cos : \mathbb{C} \rightarrow \mathbb{C}$ are entire, unique, and satisfy:

- I) $(\sin z)' = \cos z;$
- II) $(\cos z)' = -\sin z;$
- III) $\sin^2 z + \cos^2 z = 1;$
- IV) $\sin(z + w) = \sin z \cos w + \sin w \cos z;$
- V) $\cos(z + w) = \cos z \cos w - \sin z \sin w;$

for all $z, w \in \mathbb{C}$.

Example

Use the definition of π and properties III), IV), V) to show that:

- $\sin \pi = 0;$
- $\sin 2\pi = 0;$
- $\cos 2\pi = 1;$
- $\sin \pi/4 = \cos \pi/4 = \frac{1}{\sqrt{2}}.$

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Example

Show that the cosine and sine function are even and odd, respectively, in the sense that

$$\text{VI) } C(-z) = C(z) \text{ for } z \in \mathbb{C};$$

$$\text{VII) } S(-z) = -S(z) \text{ for } z \in \mathbb{C}.$$

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Remark

Unlike the real sine function, the complex sine extension is not bounded in modulus by 1, and the same is true for the complex extension of the cosine function, in fact, their moduli are unbounded.

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Theorem (Euler's Formula)

The equation

$$e^{i\theta} = \cos \theta + i \sin \theta$$

*is called **Euler's equation** (or **Euler's formula**) and holds for all $\theta \in \mathbb{R}$.*

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Example

*Prove the celebrated **Euler's identity***

$$e^{i\pi} + 1 = 0.$$

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Theorem

Given any $z \in \mathbb{C}$ with $z \neq 0$, there exists unique $r > 0$ and $w \in \mathbb{C}$, with $|w| = 1$, such that $z = rw$.

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Theorem

Let $w \in \mathbb{C}$ be such that $|w| = 1$. Then there exists a value $\theta \in \mathbb{R}$ such that $-\pi < \theta \leq \pi$ and $w = \exp(i\theta)$.

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Theorem

$\exp(z) = 1$ if and only if $z = 2k\pi i$ for some $k \in \mathbb{Z}$.

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Corollary

Let $z \in \mathbb{C}$, $z \neq 0$. Then there exist a unique $r > 0$, and a unique $\theta \in (-\pi, \pi]$ such that $z = r \exp(i\theta)$. In fact, $r = |z|$.

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Definition

If $z \in \mathbb{C}$ and if $z \neq 0$, then the unique value $\theta \in (-\pi, \pi]$ for which $z = r \exp(i\theta)$ is called the **principal argument** of z and is denoted by $\text{Arg } z$. Any other value $\theta' \notin (-\pi, \pi]$ for which $z = r \exp(i\theta')$ is called an **argument** of z and denoted by $\arg z$.

Definition

If a complex number $z \neq 0$ is written in the form

$$z = r \exp(i\theta), \quad r = |z|, \quad \theta = \operatorname{Arg} z,$$

then we say that z is in its **polar representation**.

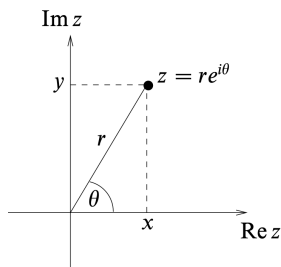


Figure 1: The polar representation of a complex number $z \neq 0$

Remark

The relations

$$|r_1 \exp(i\theta_1)r_2 \exp(i\theta_2)| = r_1 r_2 \quad \text{and} \quad \arg(r_1 \exp(i\theta_1)r_2 \exp(i\theta_2)) = \theta_1 + \theta_2$$

hold true for every pair of complex numbers $z_1 = r_1 \exp(i\theta_1)$, $z_2 = r_2 \exp(i\theta_2)$.

Thus we identify the multiplication by a complex number $z = r \exp(i\theta)$ as a rotation (counterclockwise by an angle θ) plus an elongation (by r).

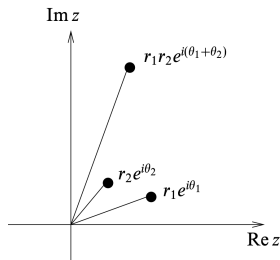


Figure 2: The polar representation of multiplication in $\mathbb{C}$

Remark

Similarly, since division can be thought of as the inverse operation of multiplication, we identify division of complex numbers as a “reverse” rotation (now clockwise by an angle θ) plus an elongation (by $1/r$).

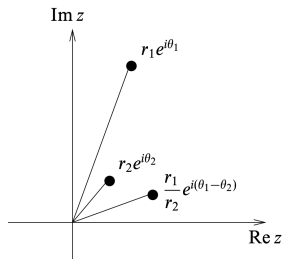


Figure 3: The polar representation of division in $\mathbb{C}$

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Theorem (De Moivre's Formula)

Let $z \in \mathbb{C}$, let $r = |z|$ and let $\theta = \arg z$, then

$$z^n = r^n (\cos n\theta + i \sin n\theta)$$

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Definition

*Each complex number ζ such that $\zeta^n = 1$ is called an **n th root of unity**. Similarly, given $z \in \mathbb{C}$, each complex number w such that $w^n = z$ is called an **n th root of z** .*

Theorem (The n th Roots of Unity)

Given $n \in \mathbb{N}$, the set of all n th roots of unity is given by

$$\mathcal{R} = \{\zeta^k\}_{k=0}^{n-1}, \quad \zeta := e^{2\pi i/n}.$$

Theorem

Given $n \in \mathbb{N}, z \in \mathbb{C}$, the set of all complex numbers w such that $w^n = z$ (namely, the n th roots of z) is given by

$$W = \{te^{i\theta/n}\zeta^k\}_{k=0}^{n-1},$$

where t is the unique positive real number such that $t^n = |z|$, θ is the principal argument of z and ζ is defined as in the previous theorem.